

EX:8WRITE THE PROGRAM FOR DEPTH-FIRST SEARCH (DFS)

AIM

To develop a Python program that implements the **Depth-First Search (DFS)** algorithm to traverse all the nodes of a graph starting from a given source node.

ALGORITHM

Step 1: Start the program.

Step 2: Create the graph using an adjacency list.

Step 3: Accept the starting node from the user.

Step 4: Mark the starting node as visited and display it.

Step 5: Recursively visit each unvisited adjacent node until all reachable nodes are visited.

Step 6: End the program after traversing all the nodes.

PROGRAM CODE:

```
graph = {}
```

```
n = int(input("Enter number of vertices: "))
```

```
for i in range(n):
```

```
    node = input("Enter vertex: ")
```

```
    neighbours = input("Enter neighbours separated by space: ").split()
```

```
    graph[node] = neighbours
```

```
start = input("Enter starting node: ")
```

```
visited = set()
```

12/7/26

```
def dfs(node):  
    if node not in visited:  
        print(node, end=" ")  
        visited.add(node)  
  
        for neighbour in graph[node]:  
            dfs(neighbour)  
  
print("\nDFS Traversal:")  
dfs(start)
```

OUTPUT:

```
===== RESTART: C:/Users/Hp/AppData/Local/Programs/Py  
Enter number of vertices: 4  
Enter vertex: A  
Enter neighbours separated by space: B C  
Enter vertex: B  
Enter neighbours separated by space: D  
Enter vertex: C  
Enter neighbours separated by space:  
Enter vertex: D  
Enter neighbours separated by space:  
Enter starting node: A  
  
Graph:  
    A  
   /\   
  B  C  
   \/   
    D  
  
DFS Traversal:  
A B D C  
|
```

RESULT:

The Python program successfully implements the **Depth-First Search (DFS)** algorithm to traverse the graph. The nodes are visited and displayed in **depth-first order** starting from the given source node.